

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal aims to automate credit card approval decisions using machine learning, boosting efficiency and accuracy for banks and financial institutions. It tackles manual review inefficiencies, reduces operational risk, and improves customer experience. Key features include an XGBoost-based risk prediction model, a Flask web application for real-time decisions, and an IBM Watson ML cloud deployment pipeline.

Project Overview	
Objective	To automate credit card approval decisions using machine learning — eliminating manual review errors, reducing processing time, and delivering instant, unbiased approval or rejection predictions based on applicant financial and demographic data.
Scope	The project covers end-to-end development: data collection from Kaggle (application_record.csv + credit_record.csv), exploratory data analysis, feature engineering, training and comparing 4 ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost), deploying the best model in a Flask web application, and providing an IBM Watson ML cloud deployment pipeline.

Problem Statement	
Description	Banks and financial institutions receive thousands of credit card applications daily. Manual review of each application is slow, inconsistent, and prone to human bias, leading to poor applicant experience and increased operational costs. Additionally, there is no self-service tool for applicants to check eligibility before formally applying.
Impact	Automating the approval process will reduce review time from days to seconds, eliminate human bias in decisions, allow compliance officers to batch-screen hundreds of applicants at once, and empower prospective customers to check their eligibility before submitting a formal application — reducing unnecessary rejections and improving overall customer satisfaction.

Proposed Solution

Approach	Using machine learning to analyze applicant financial and demographic profiles and predict credit card approval likelihood. The system derives a binary risk label from historical payment STATUS codes, trains 4 ML classifiers, selects the best model (XGBoost), and deploys it via a Flask web application for real-time, instant approval predictions.
Key Features	- XGBoost-based credit risk assessment model (78.9% accuracy, 62% recall for risky class) - Real-time single applicant prediction via Flask web form - Batch CSV screening for compliance officers to screen multiple applicants at once - Feature engineering: AGE_YEARS, YEARS_EMPLOYED, IS_UNEMPLOYED derived features - IBM Watson Machine Learning deployment pipeline for cloud-hosted predictions - Class imbalance handling using class_weight='balanced' in all 4 trained models

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU, number of cores	Standard CPU (Python 3.14) GPU optional for XGBoost
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for data, models, logs	~2 GB (CSVs + PKL files)
Software		
Frameworks	Python web framework	Flask 3.x
Libraries	ML and data libraries	scikit-learn, XGBoost, pandas, numpy, matplotlib, seaborn, joblib
Development Environment	IDE / Notebook	Visual Studio Code + Jupyter Notebook extension
Cloud Platform	ML deployment service	IBM Watson Machine Learning (IBM Cloud)
Data		
Dataset	Source, size, format	Kaggle — Credit Card Approval Prediction application_record.csv (438,557 rows) credit_record.csv (1,048,575 rows)